```
    if msg.is_multipart():
        multipart_payloads(columns,msg)
    else:
        columns['content:text'] = msg.get_payload()
    table.put(row_id, columns)

connection = happybase.Connection('h-mstr')
table = connection.table('emails')
mbox=mailbox.mbox(sys.argv[1])
for message in mbox:
    store(table,message)
```

Run '*load_mbox_file.py*' with the mbox filename as a parameter. The data should be stored in the HBase table 'emails'.

In the above example, attachments have been treated like any other content. Hence, the column family 'attachment' has been ignored.

Searching the emails

There are quite a few options for SQL queries on HBase tables, including Apache Phoenix, Impala, Hbase with Hive, Apache Drill, etc. Apache Drill is interesting as it seems to be very versatile and flexible. However, at present, it only works with HBase 0.94, while the current version is 0.98. So, let us experiment with Apache Phoenix instead *(http://phoenix.apache.org/)*, which is an SQL layer specifically meant for HBase.

Installation involves copying the Phoenix-*.*jar* files in the HBase *lib* directory and restarting HBase. Phoenix comes with a Python utility, *sqlline.py*. You may create and populate a table in the utility using the standard SQL syntax and it will create an HBase table and populate it.

You might want to use the existing HBase table and emails, by creating a view of it. For example:

```
[fedora@h-mstr phoenix-4.2.2-bin]$ bin/sqlline.py
localhost
0: jdbc:phoenix:localhost> CREATE VIEW eview (pk VARCHAR
PRIMARY KEY, "envelope"."Subject" VARCHAR, "envelope"."To"
VARCHAR, "envelope"."From" VARCHAR, "envelope"."Date"
VARCHAR,"header"."Content-Type" VARCHAR) AS SELECT * FROM
"emails";
```

The double quotes are needed as *sqlline.py* assumes that the table and column names are in upper case. Column names of the view will be the qualifiers of the column families.

Now, you can run some queries on it:

```
0: jdbc:phoenix:localhost> select * from eview where
"Subject" is not null limit 5;
0: jdbc:phoenix:localhost> select * from eview where
"Content-Type" ilike '%multipart%' limit 5;
0: jdbc:phoenix:localhost> select "From", "Subject" from
eview where "Date" ilike '%jan%2011%'  limit 5;
```

In case you need to examine some other column in a column family, create another view. You can find out more about the options available at *http://phoenix.apache.org/language/*; in particular, you can learn to create secondary indices on a view.

It would be easy to include the storage of text messages, other than emails, as well in the above system. You could probably still index the contents and attachments using *elasticsearch* or a similar tool. Combining the text search with the SQL query layer on HBase makes it possible to offer remarkable solutions to your users, helping them to analyse and search massive volumes of data with minimal effort. END

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The author has earned the right to do what interests him. You can find him online at *http://sethanil.com and http://sethanil.blogspot.com*, and reach him via email at *anil@sethanil.com*